

1. Which commands will he use to complete the given task with the help of the following operation?

- Terminating Processes via Terminal

To reduce system load, Adam can use these specific commands:

* kill a single process by PID

• Command : kill <PID>

• Example : kill 1234

* kill a process by name

If Adam knows the name but not the ID, he can use pkill.

• Command : pkill <process-name>

• Example : pkill chrome.

* kill processes by name

The killall command is used to kill all instances of a process running with a specific name.

• Command : killall <process-name>

2. Write a program for process creation using C.

→ Zombie Process

Create when the child process terminates but the parent process does not call wait() the child

remain in the zombie state until the parent exist

- `nano zombie.c` = To create the zombie process program.
- `gcc zombie.c -o zombie` → compile zombie program
- `ps -el | grep z` → To check the zombie process status

Orphan Process

Creates when the parent process terminates before the child process. the child process continue execution and it adopted by the system.

- `nano orphan.c` → To create and write a C program for orphan process.
- `gcc orphan.c -o orphan` → To compile the C program
- `./orphan` → To execute the compiled program.

3. Create the process using `fork()` system call

- child Process creation
- Parent Process creation
- PPID and PID

- Process Creating using `fork()`

Return Value of <code>fork()</code>	Meaning
1. Negative value	creation of child process was unsuccessful.

zero(0)

Returned to the newly created child process.

Positive Value

Returned to the parent process.

* PID (Process ID) :-

A unique identifier for a specific process.

* PPID (Parent Process ID) :-

The ID of the process that created it.

Commands :

getpid() : Returns the PID of the current process

getppid() :- Returns the PID of the parent process

Command →

Purpose

- ps -ef → Lists all running processes with their PID and PPID.
- ps tree → show processes in a visual tree format
- top → provides a real time view of ^{running} process
- kill -9 <PID> → forcefully terminates a process using its PID.